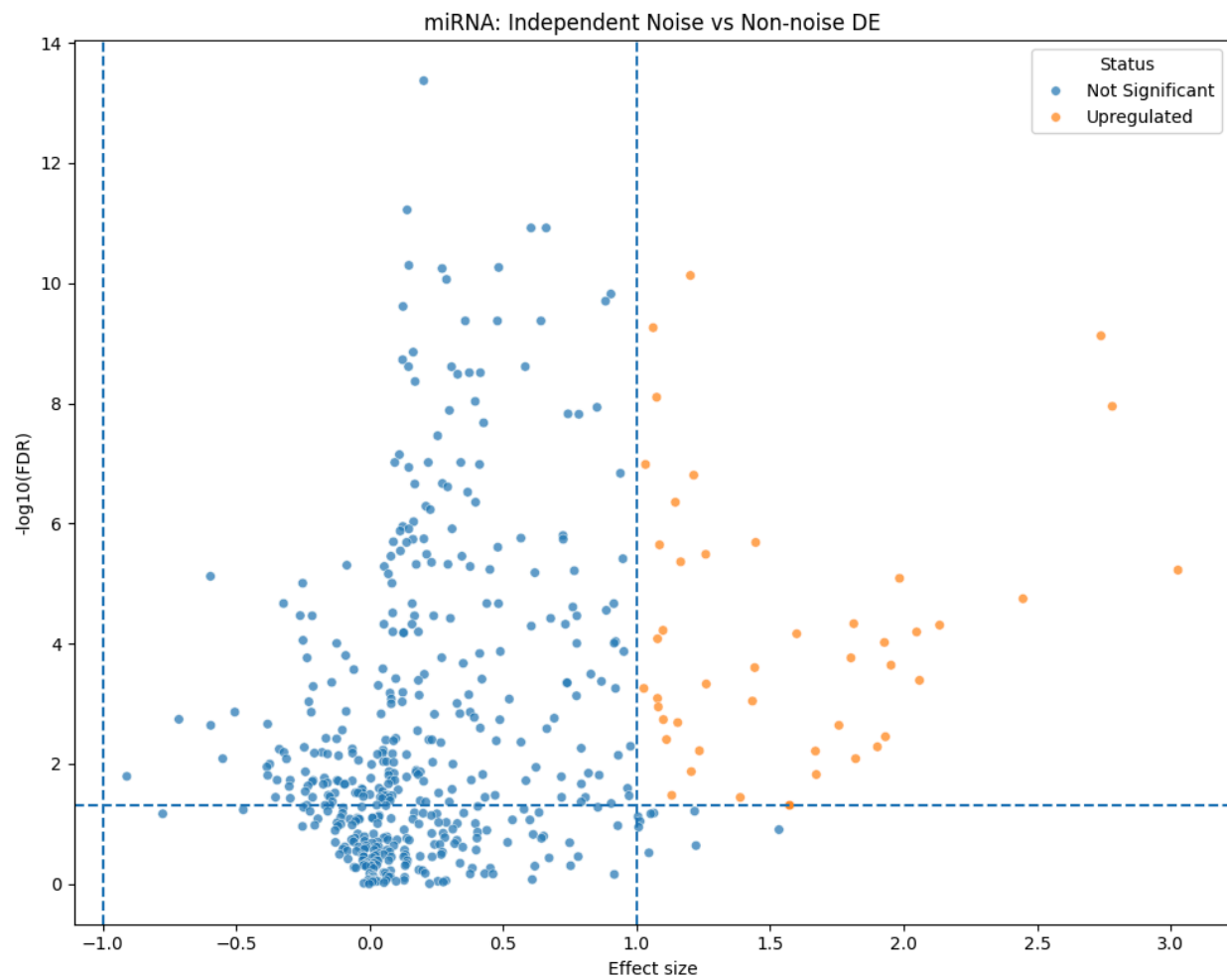
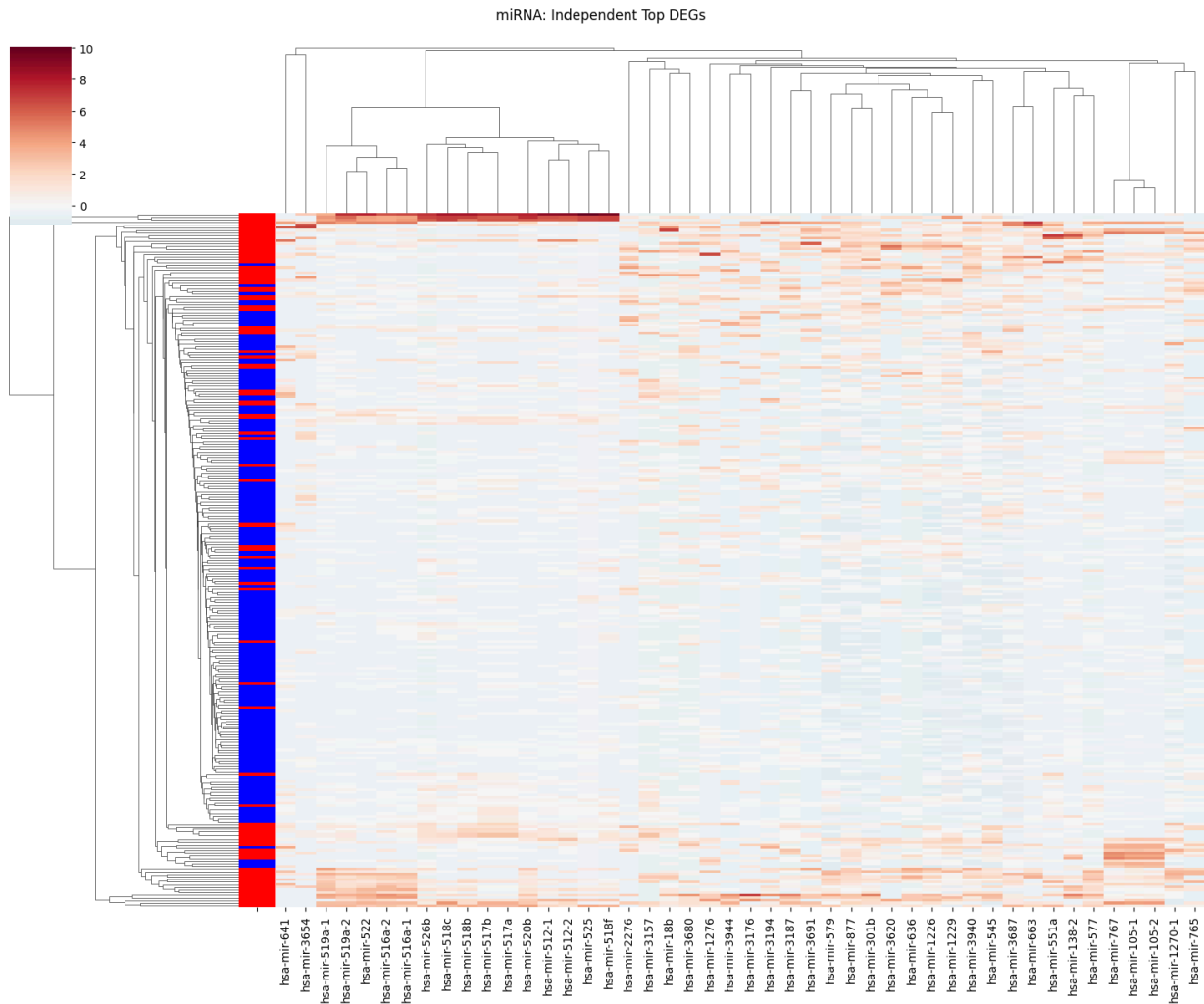






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KNOWN CANCER-ASSOCIATED miRNA ANNOTATION

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Number of annotated records: 83

Unique miRNAs: 35

Biological categories: 20

Top annotated miRNAs:

Gene	U_Statistic	P_Value	FDR	Effect_Size	Status
Known_Biological_Category	Known_miRNA				
hsa-mir-877	11934.5	1.186746e-12	7.461666e-11	1.201513	Upregulated
Apoptosis	MIR877				
hsa-mir-877	11934.5	1.186746e-12	7.461666e-11	1.201513	Upregulated
Tumor_Suppressor_miRNAs	MIR877				

hsa-mir-877	11934.5	1.186746e-12	7.461666e-11	1.201513	Upregulated
DNA_Damage_DNA_Repair	MIR877				
hsa-mir-877	11934.5	1.186746e-12	7.461666e-11	1.201513	Upregulated
Cancer_Biomarkers	MIR877				
hsa-mir-877	11934.5	1.186746e-12	7.461666e-11	1.201513	Upregulated
Study_DEG_Identified_miRNAs	MIR877				
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
Breast_Cancer_Proliferation	MIR520B				
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
Breast_Cancer_Associated	MIR520B				
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
OncomiRs	MIR520B				
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
Breast_Ovarian_Cancer	MIR520B				
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
Cancer_Biomarkers	MIR520B				
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
Study_DEG_Identified_miRNAs	MIR520B				
hsa-mir-3940	11427.5	4.257214e-10	7.931032e-09	1.075198	Upregulated
Study_DEG_Identified_miRNAs	MIR3940				
hsa-mir-519a-2	11063.5	6.478681e-10	1.123716e-08	2.781573	Upregulated
Breast_Ovarian_Cancer	MIR519A-2				
hsa-mir-519a-2	11063.5	6.478681e-10	1.123716e-08	2.781573	Upregulated
Cancer_Biomarkers	MIR519A-2				
hsa-mir-519a-2	11063.5	6.478681e-10	1.123716e-08	2.781573	Upregulated
Breast_Cancer_Associated	MIR519A-2				
hsa-mir-519a-2	11063.5	6.478681e-10	1.123716e-08	2.781573	Upregulated
OncomiRs	MIR519A-2				
hsa-mir-519a-2	11063.5	6.478681e-10	1.123716e-08	2.781573	Upregulated
Study_DEG_Identified_miRNAs	MIR519A-2				
hsa-mir-1229	11059.0	1.373229e-08	1.569851e-07	1.214133	Upregulated
Study_DEG_Identified_miRNAs	MIR1229				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
MYC_Signaling	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
Cancer_Stem_Cell	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
TGF_Beta_Signaling	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
Study_DEG_Identified_miRNAs	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
EMT_Regulation	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
Breast_Cancer_Proliferation	MIR301B				

hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
EMT_Invasion_Metastasis	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
JAK_STAT_Signaling	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
HER2_ERBB_Signaling	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
Breast_Cancer_Associated	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
Cell_Cycle_Proliferation	MIR301B				
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
PI3K_AKT_mTOR	MIR301B				

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STUDY-IDENTIFIED DIFFERENTIALLY EXPRESSED miRNAs

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Number of study miRNAs in DEG table: 35

Detected study miRNAs:

Gene	U_Statistic	P_Value	FDR	Effect_Size	Status
hsa-mir-877	11934.5	1.186746e-12	7.461666e-11	1.201513	Upregulated
hsa-mir-520b	11231.5	2.543363e-11	7.525363e-10	2.739359	Upregulated
hsa-mir-3940	11427.5	4.257214e-10	7.931032e-09	1.075198	Upregulated
hsa-mir-519a-2	11063.5	6.478681e-10	1.123716e-08	2.781573	Upregulated
hsa-mir-1229	11059.0	1.373229e-08	1.569851e-07	1.214133	Upregulated
hsa-mir-301b	10978.0	4.350911e-08	4.432738e-07	1.145205	Upregulated
hsa-mir-3187	10695.0	2.638629e-07	2.070308e-06	1.446101	Upregulated
hsa-mir-18b	10664.5	4.479621e-07	3.260715e-06	1.258980	Upregulated
hsa-mir-3687	10669.5	6.391087e-07	4.344212e-06	1.165160	Upregulated
hsa-mir-522	10129.0	9.734724e-07	5.971422e-06	3.027898	Upregulated
hsa-mir-663	10235.0	1.415141e-06	8.181794e-06	1.984873	Upregulated
hsa-mir-516a-2	10262.0	3.200467e-06	1.788706e-05	2.447343	Upregulated
hsa-mir-105-1	10156.0	9.817124e-06	4.658503e-05	1.813456	Upregulated
hsa-mir-516a-1	10172.5	1.075680e-05	4.918790e-05	2.135023	Upregulated
hsa-mir-579	10285.5	1.337524e-05	6.006916e-05	1.098974	Upregulated
hsa-mir-512-1	10100.0	1.461409e-05	6.392077e-05	2.048919	Upregulated
hsa-mir-1276	9928.0	1.608150e-05	6.855081e-05	1.599756	Upregulated
hsa-mir-3680	10142.5	1.965819e-05	8.309301e-05	1.078631	Upregulated
hsa-mir-105-2	9894.5	2.319767e-05	9.564287e-05	1.928206	Upregulated
hsa-mir-138-2	10024.5	4.505830e-05	1.726275e-04	1.802887	Upregulated
hsa-mir-519a-1	10047.0	6.087796e-05	2.285195e-04	1.952907	Upregulated
hsa-mir-577	10105.5	6.735758e-05	2.509693e-04	1.442452	Upregulated

hsa-mir-512-2	9884.0	1.153814e-04	4.086532e-04	2.059656	Upregulated
hsa-mir-636	9837.0	1.383274e-04	4.701262e-04	1.260929	Upregulated
hsa-mir-3157	9857.0	1.679260e-04	5.557023e-04	1.027237	Upregulated
hsa-mir-3691	9713.0	2.889386e-04	9.027088e-04	1.433999	Upregulated
hsa-mir-3194	9693.5	3.728696e-04	1.129840e-03	1.081405	Upregulated
hsa-mir-3944	9557.0	6.464718e-04	1.837149e-03	1.100351	Upregulated
hsa-mir-2276	9439.5	7.327465e-04	2.059059e-03	1.154169	Upregulated
hsa-mir-551a	9535.0	8.268243e-04	2.290745e-03	1.757852	Upregulated
hsa-mir-525	9397.0	1.316329e-03	3.540714e-03	1.932082	Upregulated
hsa-mir-3620	9344.0	2.485187e-03	6.068198e-03	1.235258	Upregulated
hsa-mir-767	9290.0	3.586158e-03	8.203643e-03	1.820355	Upregulated
hsa-mir-641	9093.5	6.272776e-03	1.348379e-02	1.204798	Upregulated
hsa-mir-3176	8937.0	1.915150e-02	3.344862e-02	1.131725	Upregulated

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ANNOTATION COVERAGE CHECK

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DEG records without a biological annotation: 11

Unannotated DEGs:

Gene	U_Statistic	P_Value	FDR	Effect_Size	Status
hsa-mir-1226	11715.5	1.755426e-11	5.518621e-10	1.062056	Upregulated
hsa-mir-545	11137.0	8.287371e-09	1.042137e-07	1.033526	Upregulated
hsa-mir-1270-1	10774.5	2.979983e-07	2.271109e-06	1.086119	Upregulated
hsa-mir-526b	9902.5	2.557233e-04	8.141063e-04	1.078516	Upregulated
hsa-mir-765	9470.0	1.498754e-03	3.946981e-03	1.112125	Upregulated
hsa-mir-518b	9436.5	2.075232e-03	5.219208e-03	1.901942	Upregulated
hsa-mir-518c	9401.5	2.527057e-03	6.140625e-03	1.669879	Upregulated
hsa-mir-518f	9176.0	7.260959e-03	1.509199e-02	1.673122	Upregulated
hsa-mir-3654	8851.5	2.163182e-02	3.626936e-02	1.388223	Upregulated
hsa-mir-517b	8932.0	3.070241e-02	4.902639e-02	1.572687	Upregulated
hsa-mir-517a	8932.0	3.070241e-02	4.902639e-02	1.573442	Upregulated

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FINAL miRNA ANNOTATION SUMMARY

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Total DEGs: 46

Study-identified miRNAs: 40

Study miRNAs detected: 35

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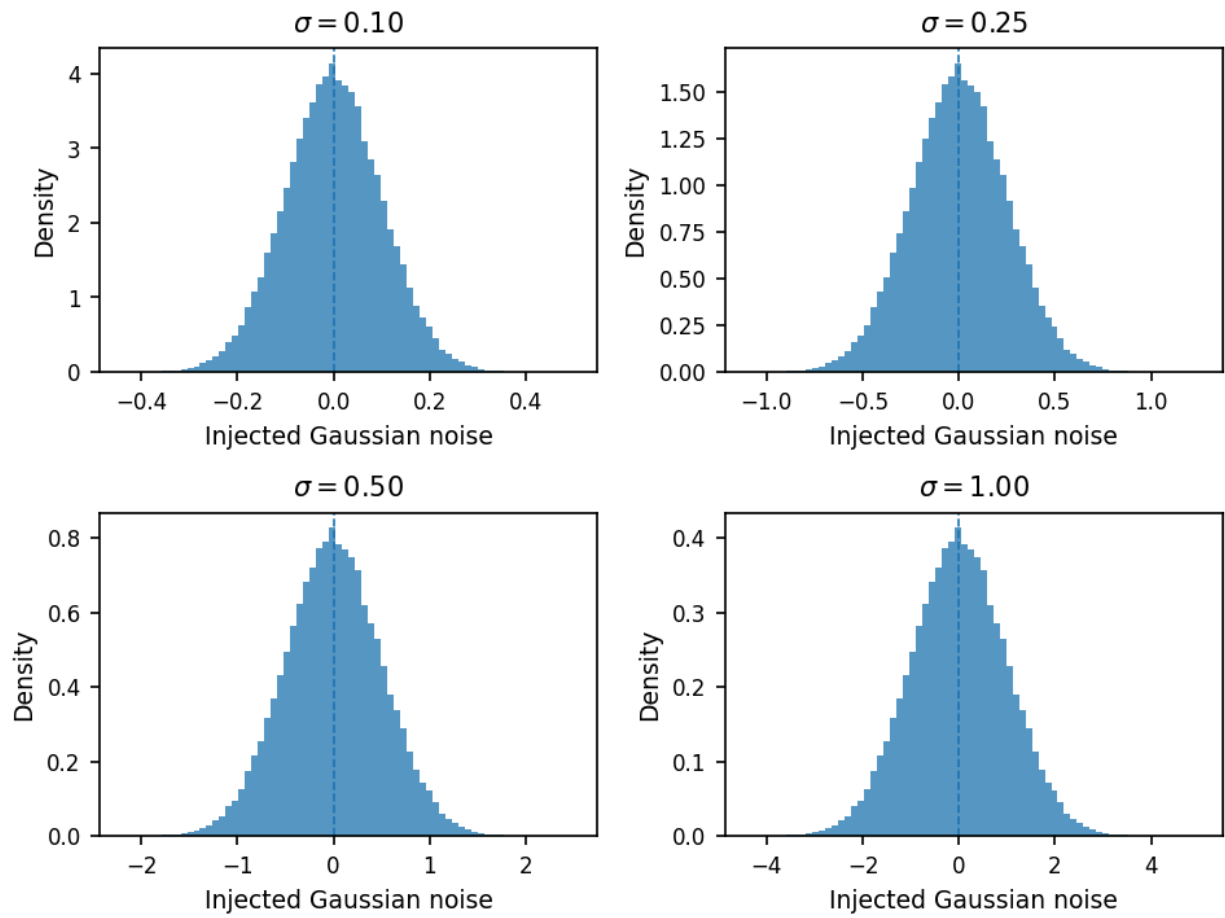
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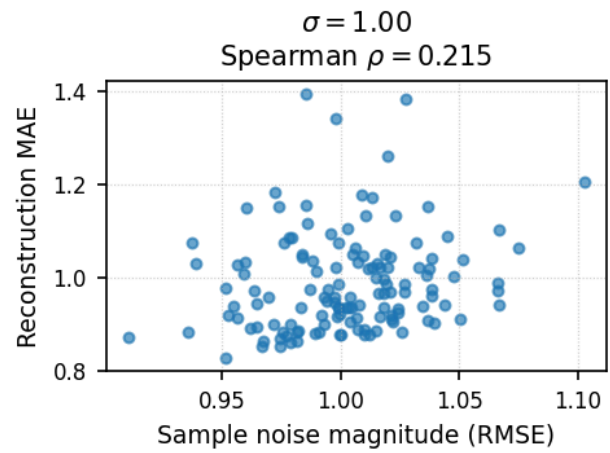
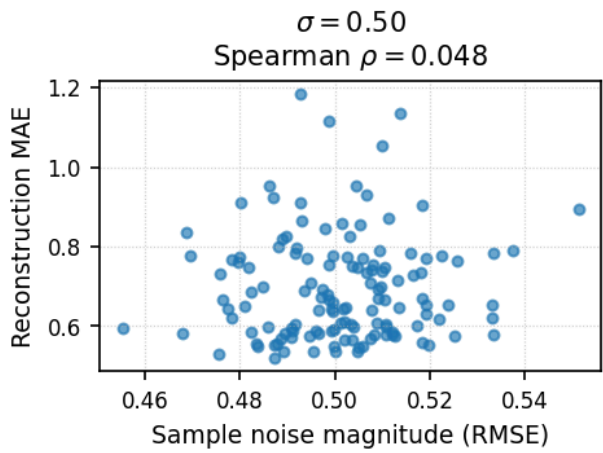
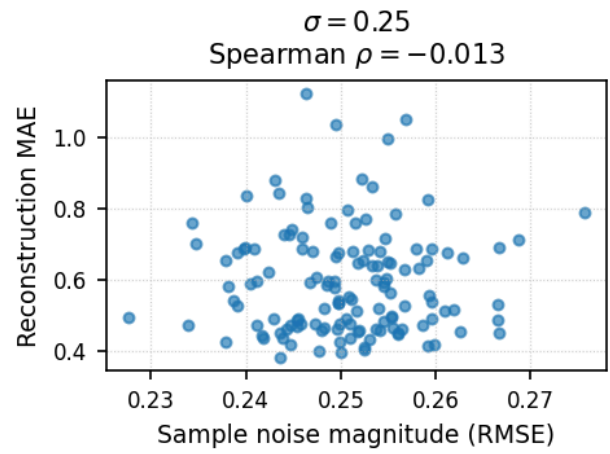
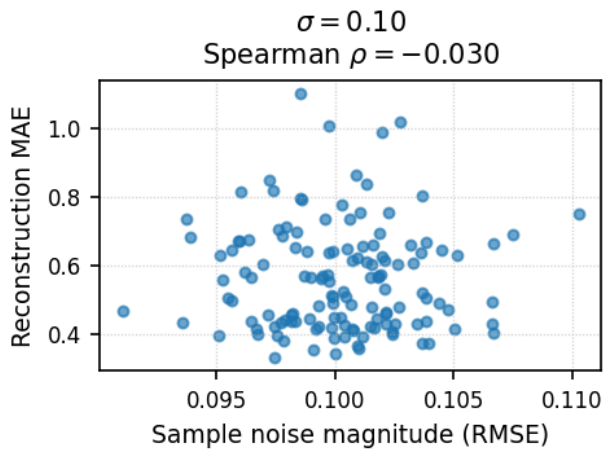
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Noise_Level	Actual_Noise_STD	Mean_MAE	Median_MAE	SD_MAE	Mean_RMSE	Median_RMSE	SD_RMSE	Max_MAE	Max_RMSE
0.10	0.100268	0.562846	0.542557	0.154586	0.748535	0.718360	0.209368	1.101337	1.442622
0.25	0.250669	0.595035	0.571071	0.148464	0.784394	0.748647	0.201725	1.122346	1.467207
0.50	0.501338	0.695723	0.666368	0.132658	0.900013	0.852169	0.180916	1.183626	1.543202
1.00	1.002676	0.989361	0.968463	0.105865	1.254244	1.221999	0.141659	1.392631	1.801441

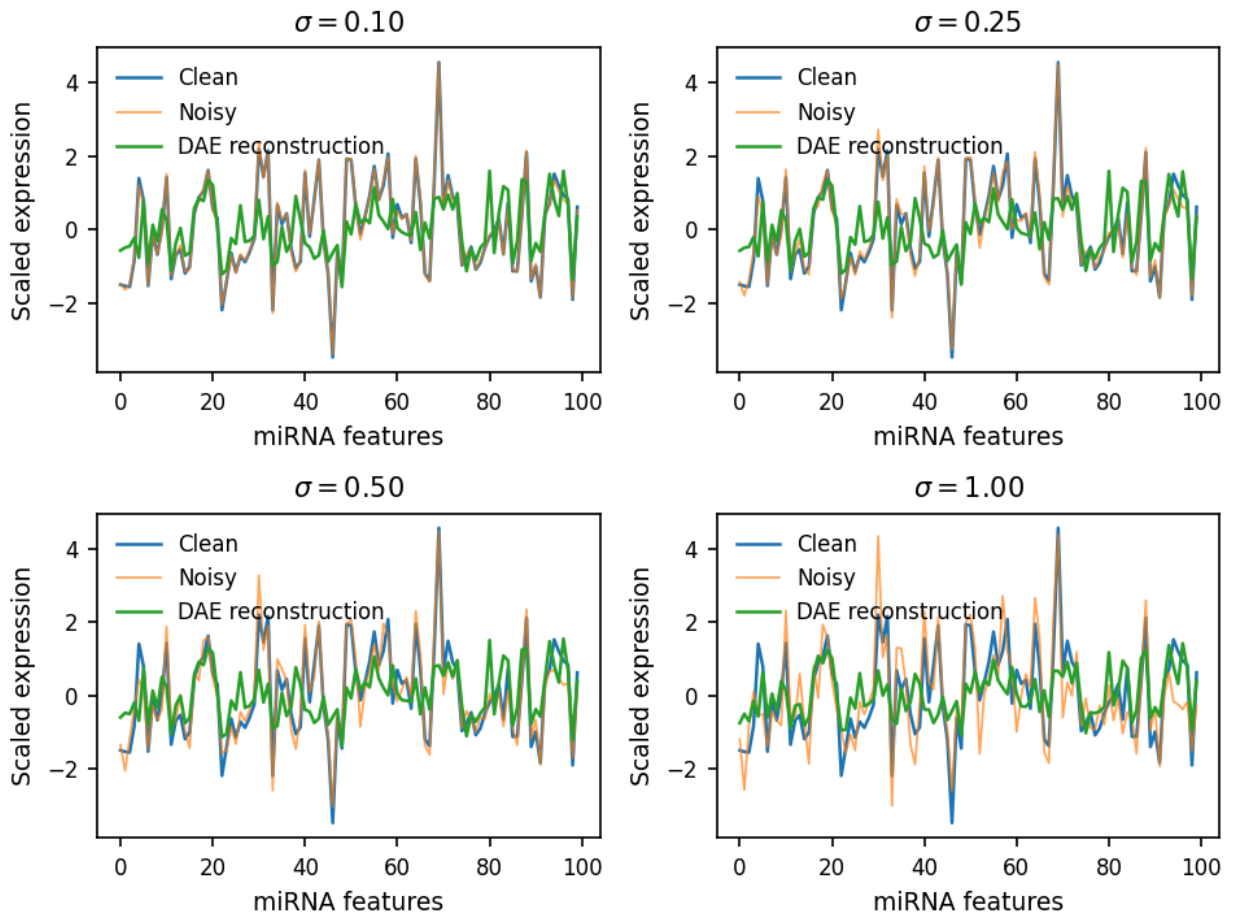
Distribution of Synthetic Technical Noise



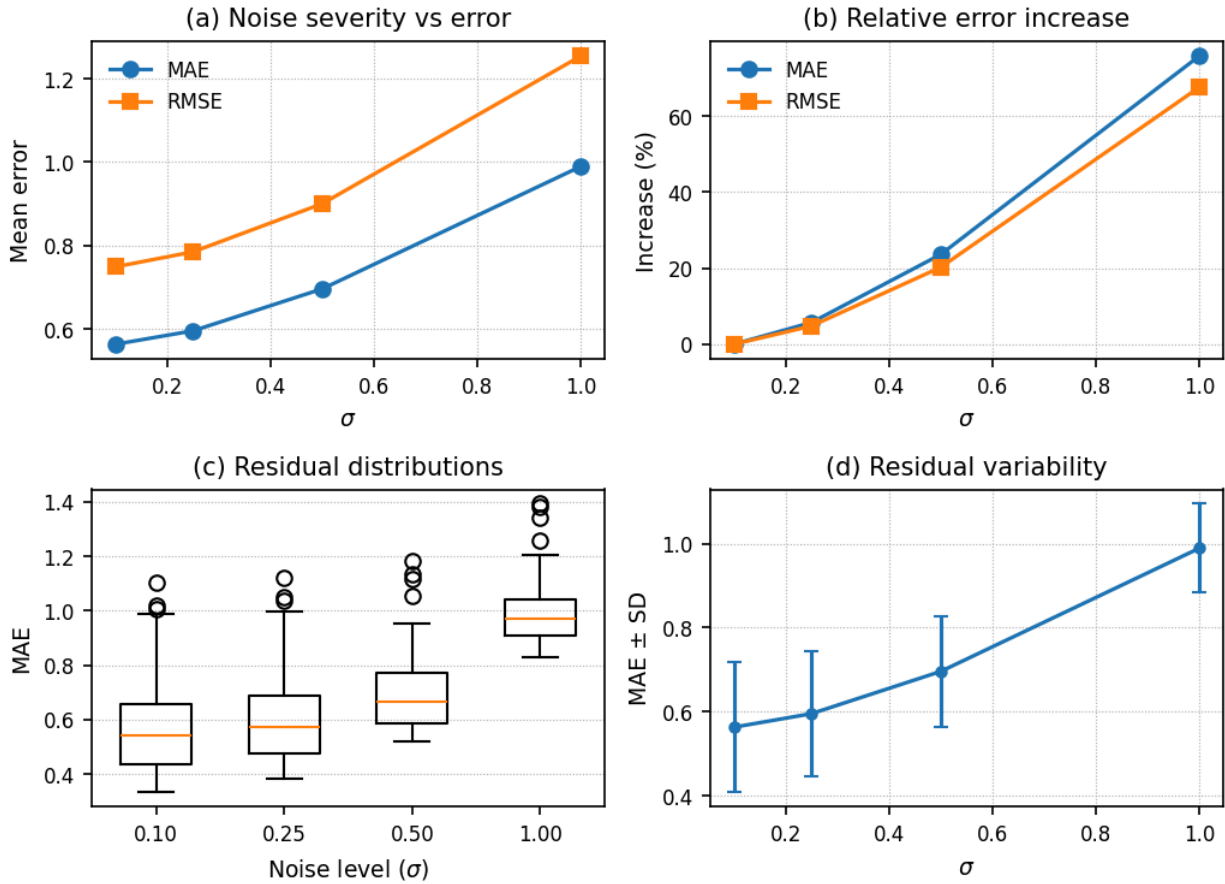
Injected Noise Magnitude vs Reconstruction Residual



Representative miRNA Signal Before and After Denoising



Synthetic Gaussian Technical-Noise Analysis of miRNA



M4 — FINAL SYNTHETIC TECHNICAL-NOISE ANALYSIS

Noise $\sigma = 0.10$

Actual noise STD : 0.100268

Mean MAE : 0.562846

Median MAE : 0.542557

Mean RMSE : 0.748535

Median RMSE : 0.718360

Noise $\sigma = 0.25$

Actual noise STD : 0.250669

Mean MAE : 0.595035

Median MAE : 0.571071

Mean RMSE : 0.784394

Median RMSE : 0.748647

Noise $\sigma = 0.50$

Actual noise STD : 0.501338

Mean MAE : 0.695723

Median MAE : 0.666368

Mean RMSE : 0.900013

Median RMSE : 0.852169

Noise $\sigma = 1.00$

Actual noise STD : 1.002676

Mean MAE : 0.989361

Median MAE : 0.968463

Mean RMSE : 1.254244

Median RMSE : 1.221999